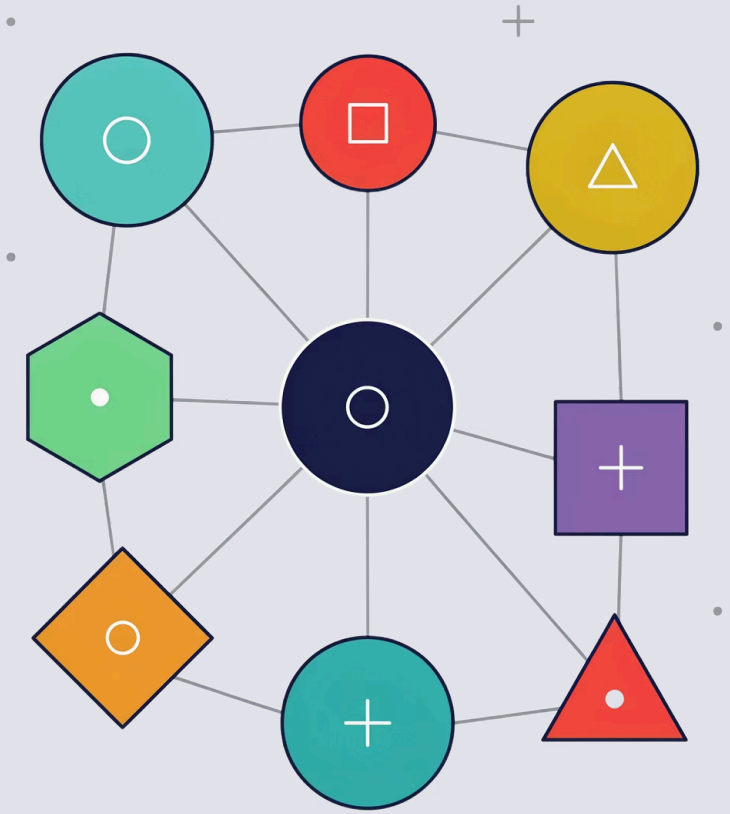
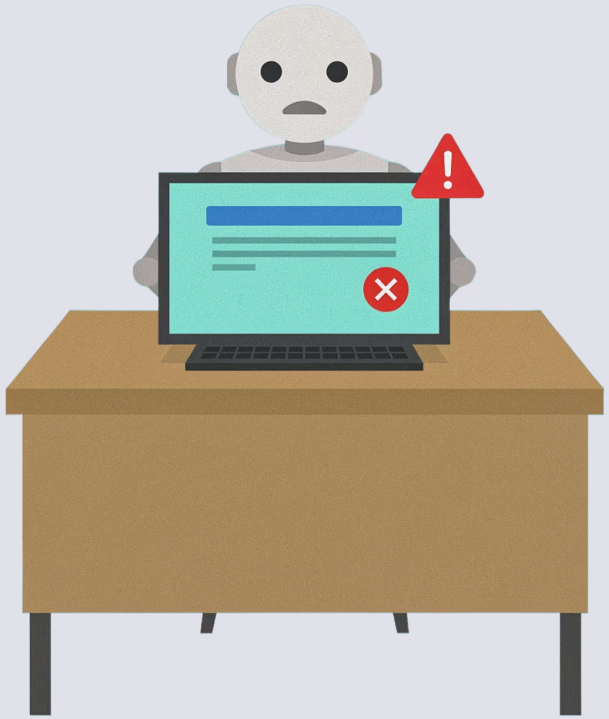


Why Does MCP Exist?

MCP addresses the limitations of LLMs when interacting with external systems

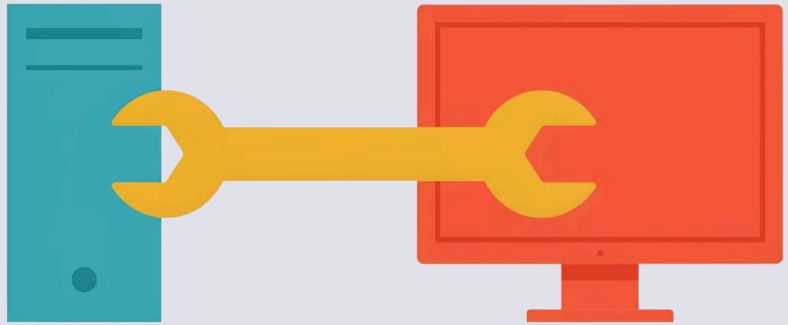


The Core Limitation of LLMs

LLMs are strong at generating content and holding conversations — writing emails, drafting campaigns, generating text

But LLMs cannot directly perform real-world actions:

- **Sending emails**
- **Running advertisements**
- **Interacting with external services**



Tool Calling as a Solution

Tool calling lets LLMs and agents interact with external systems

An **agent** = an LLM combined with a set of tools

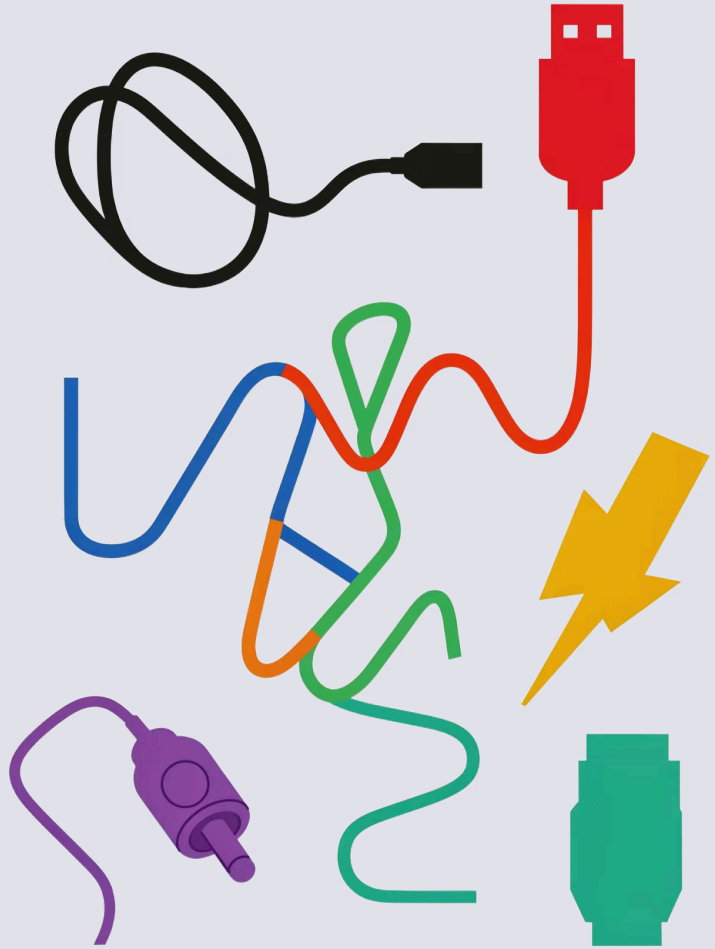
The Problem — No Standard for Tool Calling

Different frameworks implement tool calling differently:

OpenAI Agents Framework uses strict Python function-based tools

LangChain uses a different mechanism entirely

No universal standard exists for how LLMs call external tools





Duplicate Development Across the Industry

Developers repeatedly build the same integrations independently

Example: countless developers each build their own Gmail API interface

Result: duplicated effort industry-wide, solving the same problem over and over

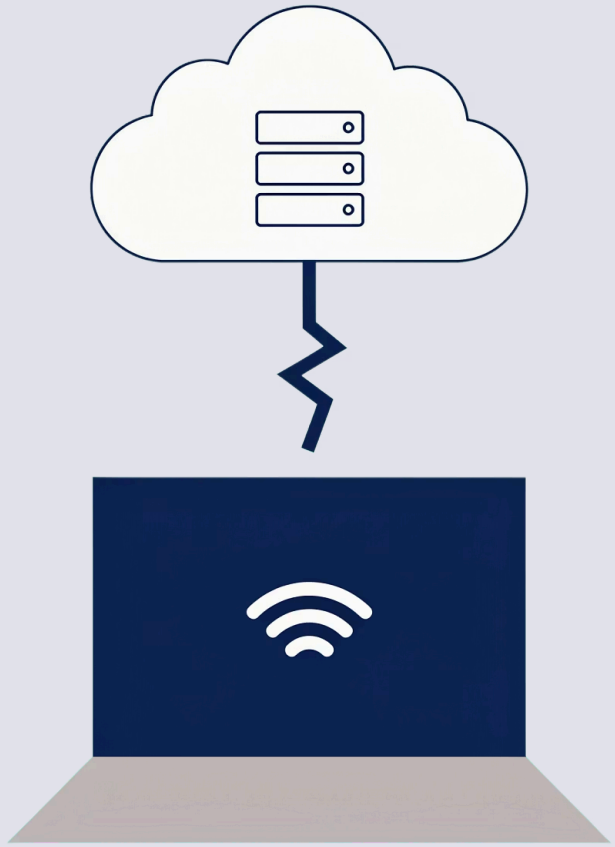
The Maintenance Burden

External APIs change over time — e.g. Gmail API updates can break existing integrations

Developers must constantly maintain and update their custom tool interfaces

Every custom integration becomes an ongoing liability, not a one-time build





The Local Access Gap

Most existing tool calls rely on remote APIs and servers — Gmail, Weather, Airbnb, and other web services

This leaves LLMs unable to easily interact with a user's local environment

What Local Integration Unlocks

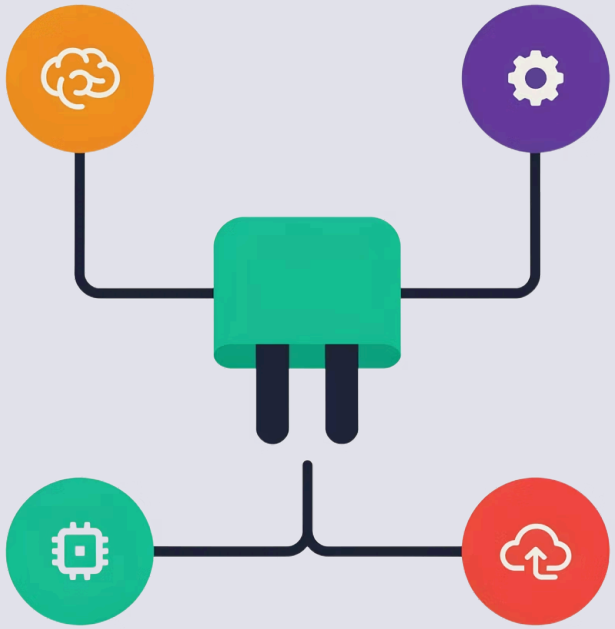
Many useful AI interactions involve local machine resources:

Organizing local files

Monitoring screen activity to suggest productivity improvements

Reading local source code in real time in editors like VS Code

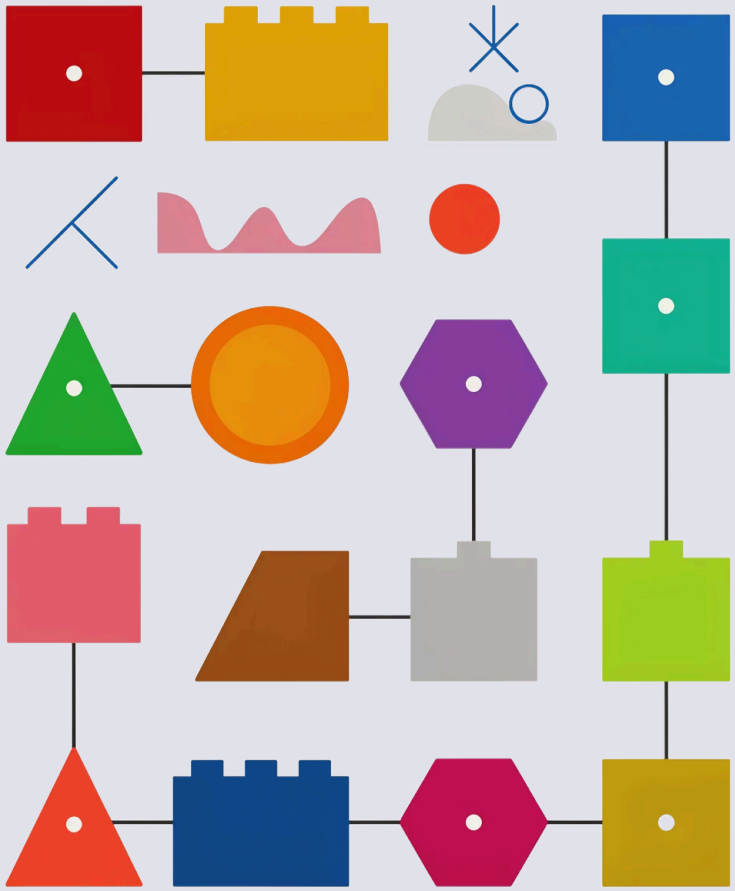




The Core Purpose of MCP

Provide a standardized way for AI systems
to interact with external tools and systems

Enable interoperability between different
AI models and external services

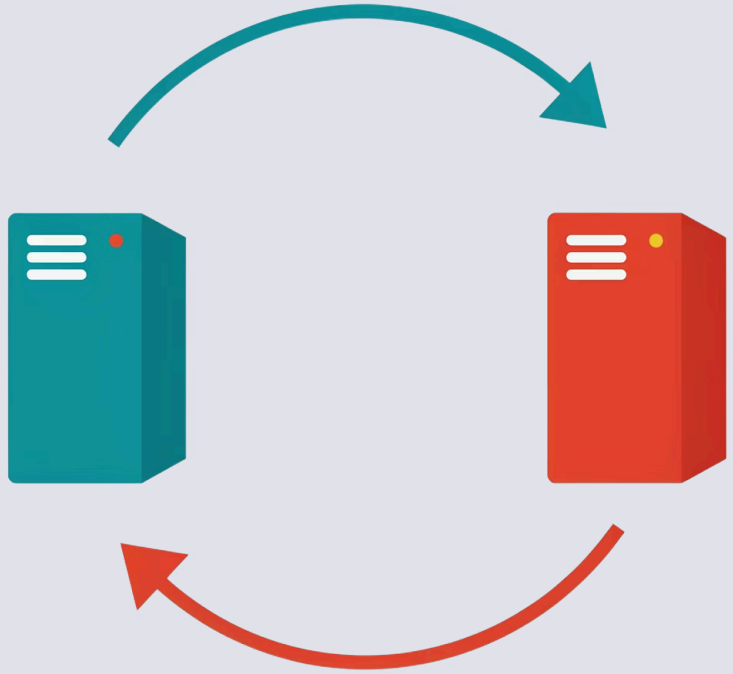


The Key Benefit — A Reusable Ecosystem

Developers build MCP servers that expose tools and resources

Any MCP client can connect to any MCP server

Creates a reusable, interoperable ecosystem for AI integrations — build once, connect anywhere



What MCP Actually Is

MCP is not a revolutionary new technology

It's a widely adopted **standard protocol** that simplifies AI integrations